DAY 1
Introduction to PDEs and Modeling

Plan for the Lecture

- Recall derivatives from analysis
- What is a partial differential equation (PDE)?
- Where do PDEs pop up?
- Examples from
  - Gravity
  - Conservation laws (mass, momentum energy)
  - Groundwater flow

Derivatives

- **Derivative** of $f : \mathbb{R} \to \mathbb{R}$ **in** $x \in \mathbb{R}$

$$\frac{df}{dx}(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Limit independent of sequence $\Delta x \to 0$

- **Partial derivative** of $f : \mathbb{R}^d \to \mathbb{R}$ w.r.t. x_i **in** $x \in \mathbb{R}^d$

$$\frac{\partial f}{dx_i}(x) = \lim_{\Delta x_i \to 0} \frac{f(\ldots, x_i + \Delta x_i, \ldots) - f(x)}{\Delta x}$$

Limit independent of sequence $\Delta x_i \to 0$

Derivatives Continued

- f is partially **differentiable** in $D \subseteq \mathbb{R}^d$ if all its partial derivatives exist for all $x \in D$ and the function

$$\partial_{x_i} f : D \to \mathbb{R}$$

given by the function values

$$\partial_{x_i} f(x) = \frac{\partial f}{\partial x_i}(x)$$

is continuous in D.

- **Note**: $\partial_{x_i} f(x)$ is a number in $\mathbb{R}$ and $\partial_{x_i} f$ is a function
- Higher-order partial derivatives

$$\partial_{x_i} \partial_{x_i} f = \partial^2_{x_i^2} f = \frac{\partial^2 f}{\partial x_i^2} = \partial^2_{x_i x_i} f$$

Nabla & Friends

- **Gradient** of $f : \mathbb{R}^d \to \mathbb{R}$

$$\nabla f = \begin{pmatrix} \partial_{x_1} f \\ \vdots \\ \partial_{x_d} f \end{pmatrix}, \quad \nabla f : \mathbb{R}^d \to \mathbb{R}^d$$

- **Divergence** of $f : \mathbb{R}^d \to \mathbb{R}^d$

$$\nabla \cdot f = \partial_{x_1} f_1 + \cdots + \partial_{x_d} f_d = \sum_{i=1}^{d} \partial_{x_i} f$$

- **Laplacian** of $f : \mathbb{R}^d \to \mathbb{R}$

$$\Delta f = \partial^2_{x_1^2} f + \cdots + \partial^2_{x_d^2} f = \sum_{i=1}^{d} \partial^2_{x_i^2} f$$

- **Jacobian** of $f : \mathbb{R}^m \to \mathbb{R}^n$

$$\nabla f = \begin{pmatrix} \partial_{x_1} f_1 & \cdots & \partial_{x_n} f_1 \\ \vdots & & \vdots \\ \partial_{x_1} f_m & \cdots & \partial_{x_n} f_m \end{pmatrix}$$

- ∇ is polymorphic
- ∇ is called **Nabla operator**
- Be careful, „·“ can make a big difference

Partial Differential Equations

- A partial differential equation (PDE)

seeks a function f in several variables
satisfying conditions on its partial derivatives

- Example: Let $f : \mathbb{R}^2 \to \mathbb{R}$ and require

$$\partial_{x_1} f + \partial_{x_2} f = g$$

  where $g : \mathbb{R}^2 \to \mathbb{R}$ is a given function

- This means, for **every** $x \in \mathbb{R}^2$ we want

$$\frac{\partial f}{\partial x_1}(x) + \frac{\partial f}{\partial x_2}(x) = g(x)$$

- Often, the PDE is posed on a bounded domain $\Omega \subset \mathbb{R}^2$. Then boundary conditions are necessary. Today we don't talk about boundary conditions.

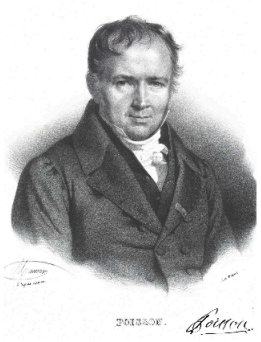
Partial Differential Equations are Everywhere!

- **Fluid Mechanics**
  - Lift and drag of airfoil, car, ship
  - Weather prediction
  - Climate prediction
  - Tsunami (flooding) prediction
  - Groundwater availability
  - Combustion in a flow
  - Contaminant transport
  - Oil production
- **Heat flow**
  - Heat distribution in a computer chip
  - Geothermal heating/cooling
  - Nuclear reactor design
- **Solid Mechanics**
  - Deformation of a bridge
  - Crack formation, e.g. in fracking
  - Movement of a biological cell
- **Wave Problems**
  - Sound waves
  - Earth quakes
  - Electromagnetic scattering
- **Quantum Mechanics**
- **General Relativity**
- **…..**

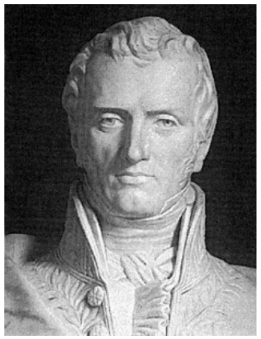
Historical Perspective

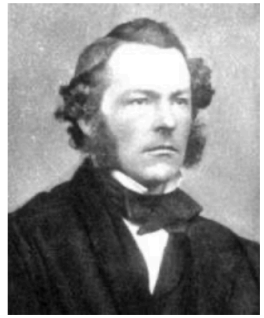
- Calculus was invented for differential equations!
- e.g. to express conservation of mass, momentum and energy in quantitative form
- Famous examples:
  - Euler (inviscid flow) 1757
  - Poisson (electrostatics, gravity) 1800
  - Navier-Stokes (viscous flow) 1822/1845
  - Maxwell (electrodynamics) 1864
  - Einstein (general relativity) 1915
  - Schrödinger (quantum mechanics) 1926
- **Why is it still important?**

  Solutions in practical situations only possible with numerical methods on modern (super-)computers!

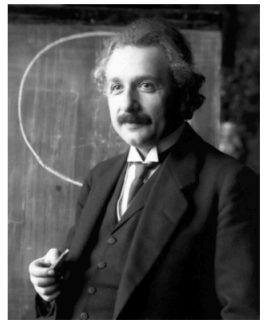

Poisson


Euler


Navier


Stokes


Maxwell


Einstein

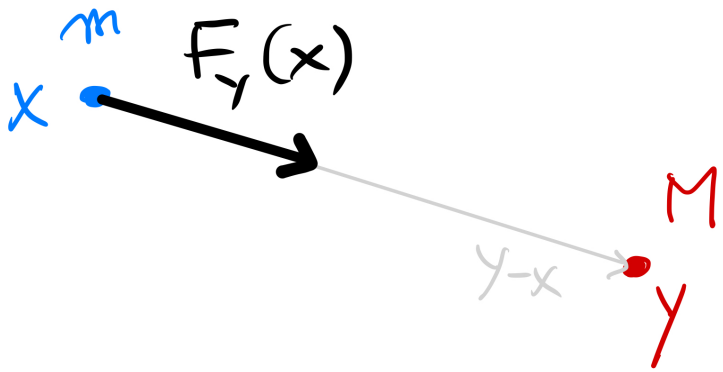
A Word on Supercomputers

- You can have a supercomputer under your desk!
- ASCI Red (historical supercomputer, ~50M$ in 1997)
  - 9152 Pentium Pro 200 MHz
  - 1212 GB RAM
  - 1 TFLOP (10^{12} floating point operations per second) Linpack performance
  - #1 TOP 500 1997-2000
- State-of-the-art workstation (~15k€):
  - 1 AMD Ryzen PRO 9975WX CPU with 32 cores
  - 256 GB RAM (1500 max)
  - 1.5 TFLOP with self-written LU factorization code
  - it also has an NVIDIA RTX 5090 GPU with 100 TFLOP peak in FP32
- **You can do a lot of cool science with that! You are only limited by ideas!**

Example: Gravity

- Newton's law of gravity: force on a mass m located at x excerted by a mass M located at y:

$$F_y(x) = \frac{\gamma m M}{\|y-x\|^2} \frac{y-x}{\|y-x\|}$$



- Holds for homogeneous, perfectly spherical masses with arbitrary diameter **outside** of the mass
- Together with Newton's second law: System of ordinary differential equations for position, velocity
- What about gravitational force
  - within a massive body?
  - within a body consisting of material with different mass densities?
  - E.g. a gas cloud in the universe?

https://esahubble.org/images/heic0206c/



Poisson's Equation

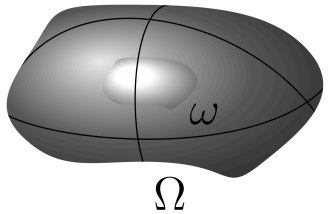
Norm: $\|x\| = \sqrt{\sum_{i=1}^{3} x_i^2}$ Gradient: $\nabla f(x) = \begin{pmatrix} \partial_{x_1} f(x) \\ \partial_{x_2} f(x) \\ \partial_{x_3} f(x) \end{pmatrix}$

- Introduce **gravitational potential**

$$\psi_y(x) = -\frac{\gamma M}{\|x-y\|}, \qquad -\nabla\psi_y(x) = \frac{\gamma M(y-x)}{\|y-x\|^3} = \frac{F_y(x)}{m} = a_y(x)$$

- **Continuum assumption**: Distributed mass is given by its mass density

$$\rho(x) : \mathbb{R}^3 \to \mathbb{R}, \qquad M_\omega = \int_\omega \rho(x)\, dx$$



- Gravitational potential $\Psi : \mathbb{R}^3 \to \mathbb{R}$ of the distributed mass is then given by

$$\Delta\Psi(x) = 4\pi\gamma\rho(x) \quad (x \in \mathbb{R}^3) \quad \textbf{Poisson's equation}$$

- and the acceleration at position x is then given by

$$a(x) = -\nabla\Psi(x)$$

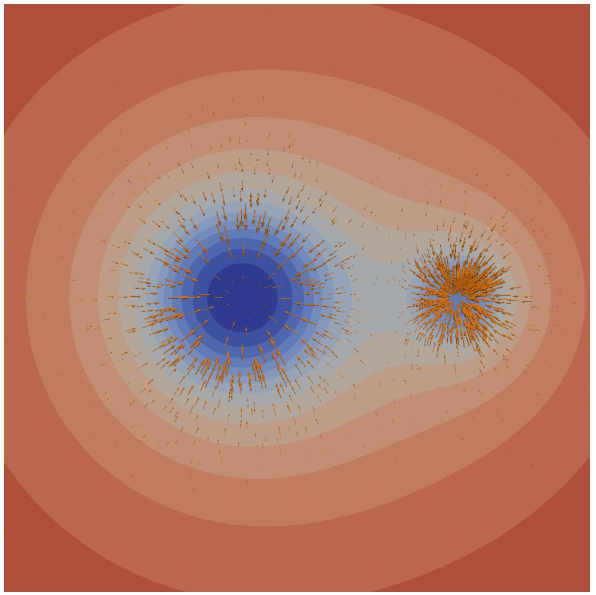
Laplacian:

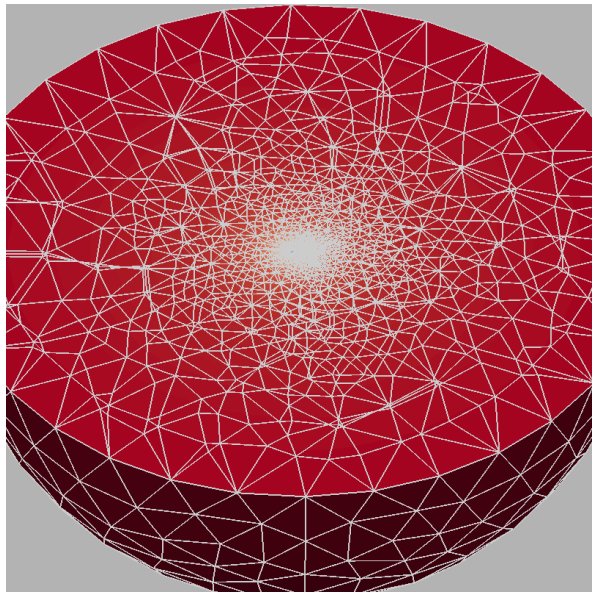
$$\Delta f(x) = \frac{\partial^2 f}{\partial x_1^2}(x) + \frac{\partial^2 f}{\partial x_2^2}(x) + \frac{\partial^2 f}{\partial x_3^2}(x)$$

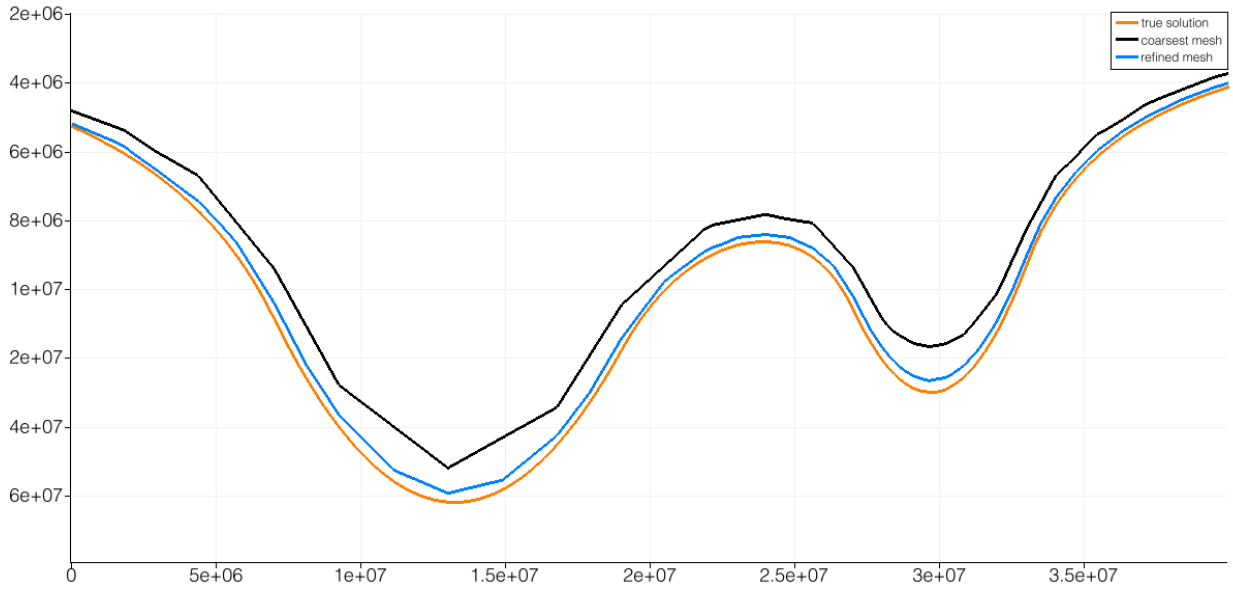
Example for Gravity

$$\Delta\Psi(x) = 4\pi\rho(x) \qquad \text{in } \Omega$$
$$u = 0 \qquad \text{on } \partial\Omega$$

$$\rho(x) = \begin{cases} \rho_p & \|x-p\| \le R_p \\ \rho_q & \|x-q\| \le R_q \\ 0 & \text{else} \end{cases}$$

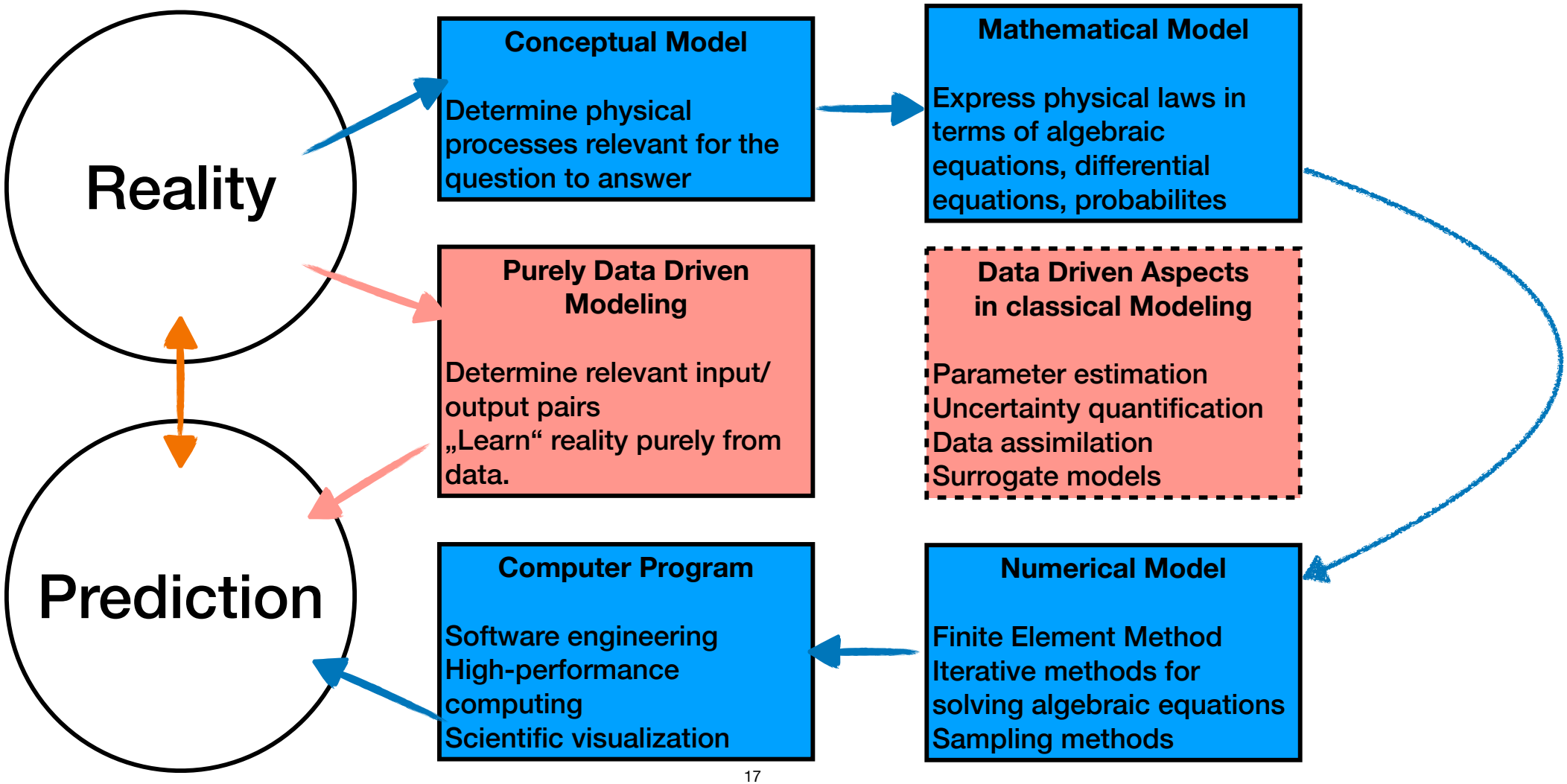






- Ω: **finite domain**
- Numerical solution needs triangulation of the domain
- Plot shows potential computed with finite element method on different meshes and the true solution

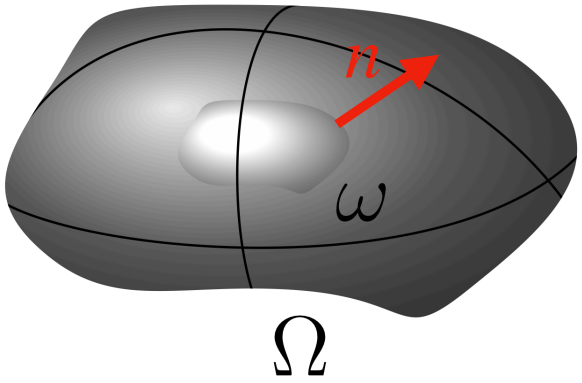
Modeling, Simulation, Optimization, Learning



Why is „Classical“ Modeling Still Important?

- Data-driven modeling constructs a function $f : X \to Y$ that can be evaluated
- Classical models based on PDEs give more insight into the input-output relation, e.g.
  - Dependence on parameters
  - Relative influence of different effects
  - asymptotic behavior
- Better generalization in data sparse regime
- Approaches should not be seen as opposites but rather be combined:
  - Learning submodels in PDE models, e.g. heat conductivity coefficent, water retention curve, turbulent viscosity
  - Constructing surrogate models using simulations as data

PDEs from Conservation Laws



- We formulate laws of physics in a **domain** $\Omega \subseteq \mathbb{R}^d$
- Domains are connected, open sets, **dimension** $d = 1,2,3$
- Recall mass density (unit: kg per m³)

$$\rho(x) : \mathbb{R}^3 \to \mathbb{R}, \qquad M_\omega = \int_\omega \rho(x)\, dx$$

- M_ω is the mass (in kg) contained in $\omega \subseteq \Omega \subset \mathbb{R}^d$, ω is fixed in time
- The **change of mass in** ω in the time interval $(t, t + \Delta t)$ is

$$M_\omega(t + \Delta t) - M_\omega(t) = \int_t^{t+\Delta t} \left\{ \int_\omega f(x, r)\, dx - \int_{\partial\omega} \rho(x, r) v(x, r) \cdot n(x)\, ds \right\} dr$$

- where: f are sources, $v(x, t)$ is a velocity field, $n(x)$ is the unit outer normal vector

Conservation of Mass (2)

- Using $\int_t^{t+\Delta t} g(r)\,dr = \Delta t\, g(t) + O(\Delta t^2)$ we obtain

$$\frac{M_\omega(t+\Delta t) - M_\omega(t)}{\Delta t} = \int_\omega f(x,t)\,dx - \int_{\partial\omega} \rho(x,r)v(x,r)\cdot n(x)\,ds + O(\Delta t)$$

- Take limit $\Delta t \to 0$, insert definition of M_ω and use Gauß theorem:

$$\frac{d}{dt}\int_\omega \rho(x,t)\,dx + \int_\omega \nabla\cdot(\rho(x,t)v(x,t))\,dx = \int_\omega f(x,t)\,dx$$

- For sufficiently smooth functions this implies the partial differential equation

$$\partial_t\rho(x,t) + \nabla\cdot(\rho(x,t)v(x,t)) = f(x,t)$$

- called **conservation of mass**

Divergence: $\nabla\cdot f(x) = \frac{\partial f_1}{\partial x_1}(x) + \frac{\partial f_2}{\partial x_2}(x) + \frac{\partial f_3}{\partial x_3}(x)$

What does it mean?

- The mass conservation equation

$$\partial_t \rho(x,t) + \nabla \cdot (\rho(x,t) v(x,t)) = f(x,t)$$

  states a condition on the three functions $\rho(x,t), v(x,t), f(x,t)$

- $f(x,t)$, the source/sink term is usually given
- If the velocity field $v(x,t)$ is also given (see below), then $\rho(x,t)$ can be computed
- If a fluid is incompressible, then its density is constant, so $\partial_t \rho(x,t) = 0$
- If in addition $f(x,t) = 0$ (no sources/sinks in Ω), then the velocity field of the fluid must satisfy

$$\nabla \cdot v(x,t) = 0$$

  for all (x,t).

Conservation of Heat Energy

- Heat energy density $e(x,t)$ (in Joule per m^3) is $e(x,t) = \rho(x,t)cT(x,t)$, where ρ is mass density, c is specific heat capacity, $T(x,t)$ is temperature (Kelvin)
- Heat energy stored in ω is $Q_\omega(t) = \int_\omega \rho(x,t)cT(x,t)\,dx$
- The argument above can be repeated and results in the **heat equation**

$$\partial_t(\rho cT) + \nabla \cdot (\rho cTv - \lambda \nabla T) + r\rho cT = f$$

- where
  - $r\rho cT$ is the **sink term** (heat is removed)
  - $q_c = \rho cTv$ is **convective heat flux**, v is again a velocity field
  - $q_d = -\lambda \nabla T$ is **diffusive heat flux** (or Fourier's law) with λ heat conductivity
  - Fourier's law says heat flows in direction of „steepest decrease of temperature“
  - f is the **source term** (heat is added)

Special Cases of the Heat Equation

- Here it is again:

$$\partial_t(\rho c T) + \nabla \cdot (\rho c T v - \lambda \nabla T) + r \rho c T = f$$

- Stationary case, no sinks

$$\nabla \cdot (\rho c T v - \lambda \nabla T) = f$$

- Stationary case, solid body (no convective flux), no sinks

$$-\nabla \cdot (\lambda \nabla T) = f$$

- Stationary case, solid body, no sinks, λ constant

$$\nabla \cdot \nabla T = \Delta T = -f/\lambda$$

This is **Poisson's equation again!**

Conservation of Linear Momentum

- Linear momentum density is $\rho(x,t)v(x,t)$ (density times velocity)
- Amount of linear momentum in ω then is the vector-valued quantity

$$P_\omega(t) = \int_\omega \rho(x,t)v(x,t)\,dx$$

- Each component then has its conservation equation

$$\partial_t(\rho(x,t)v_i(x,t)) + \nabla \cdot j_i = f_i(x,t)$$

  with $j_i = \rho v_i v$, for *inviscid flow* (momentum is only transported „with the flow“)
- This results in a system of three coupled partial differential equations
- Defining the matrix $J = \rho v v^T$ and divergence of a matrix appropriately gives

$$\partial_t(\rho(x,t)v(x,t)) + \nabla \cdot \left(\rho v v^T\right) = f(x,t)$$

Inviscid Fluid Flow

- The famous **Euler equations of gas dynamics** are given by

$$\partial_t \rho + \nabla \cdot (\rho v) = m$$
$$\partial_t (\rho v) + \nabla \cdot (\rho v v^T + pI) = f$$
$$\partial_t e + \nabla \cdot ((e + p)v) = s$$

- consisting of
  - Conservation of mass
  - Conservation of momentum
  - Conservation of energy

  for the unknown functions ρ (mass density), v (velocity), e (energy density) and p pressure.
- Inviscid means that internal friction in the fluid is neglected
- It needs to be closed with a thermodynamical relation (polytropic ideal gas)

$$p = p(\rho, e) = (\gamma - 1)(e - \rho \|v\|^2/2)$$

Propagation of Sound Waves

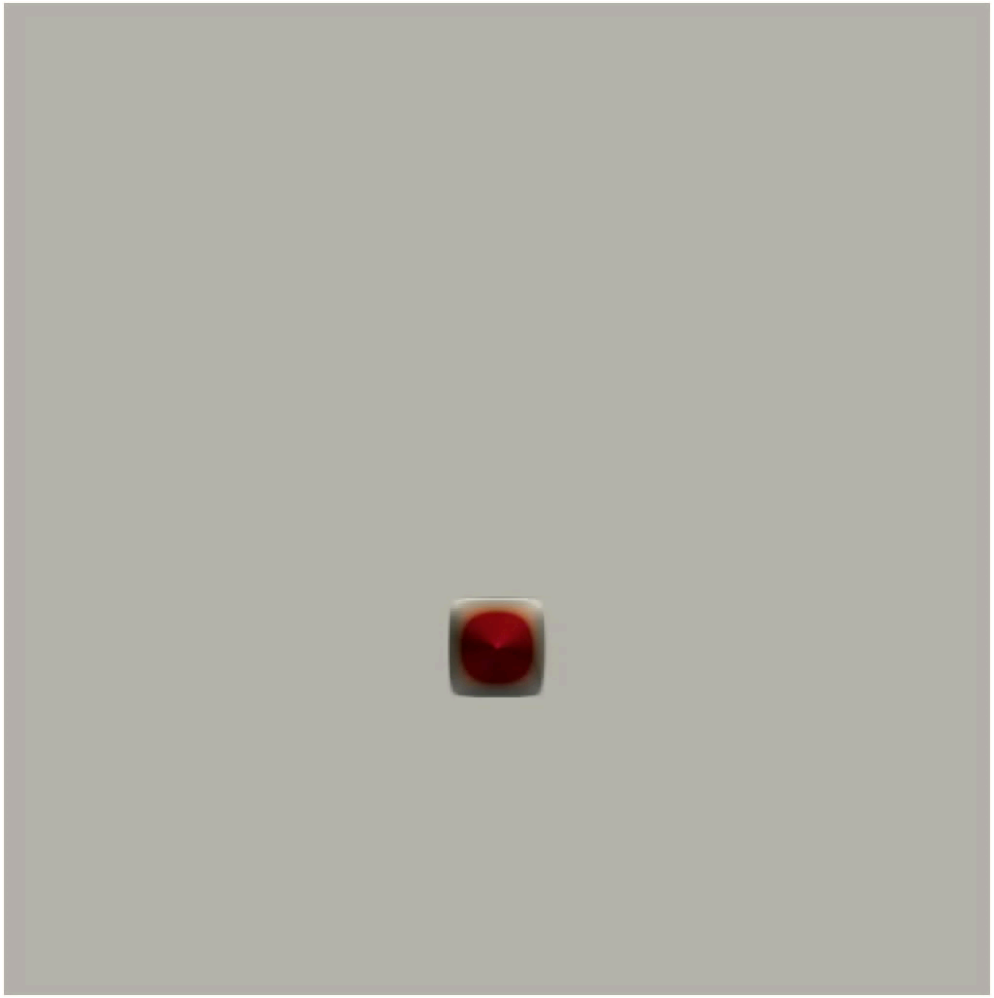
- Sound waves are small variations in pressure (and correspondingly density) that move through a gas
- Linearizing the Euler equations around a constant state using

$$\rho = \bar{\rho} + \tilde{\rho}, \qquad p = \bar{p} + \tilde{p}, \qquad v = \bar{v} + \tilde{v}$$

- and dropping higher order terms results in the linear acoustics equations

$$\partial_t \tilde{\rho} + \nabla \cdot (\bar{\rho}\tilde{v}) = 0$$

$$\partial_t (\bar{\rho}\tilde{v}) + \nabla \tilde{p} = 0$$



Incompressible Navier-Stokes Equations

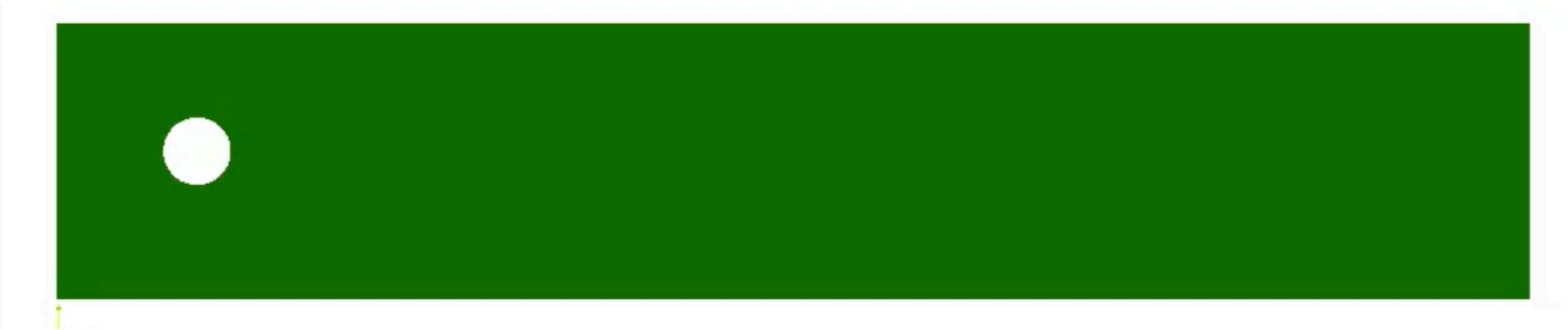
- In this flow model internal friction is considered („Newtonian fluid“)
- For an incompressible fluid, the model reads

$$\nabla \cdot v = 0$$

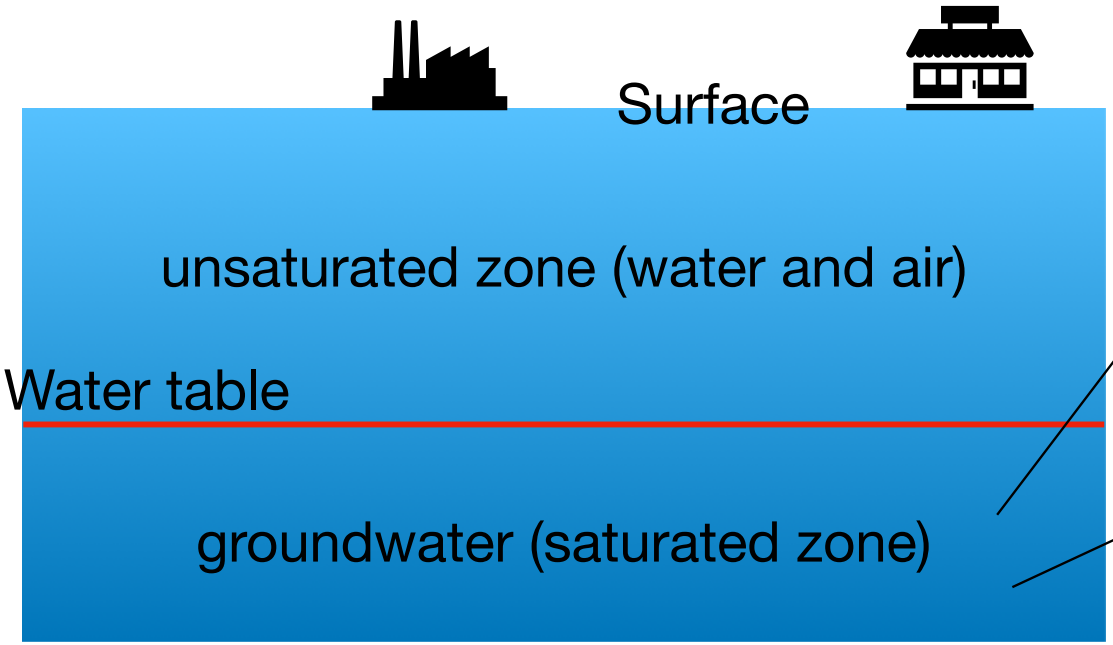
$$\partial_t v + \nabla \cdot (vv^T) - \kappa \Delta v + \nabla p = f$$

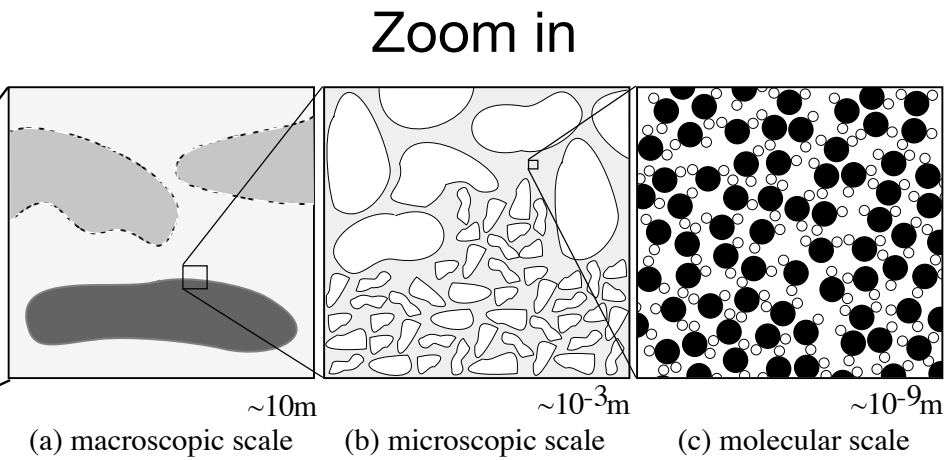
  where κ is the *kinematic viscosity* of the fluid

- The first equation is conservation of mass, the second conservation of momentum (von Karmann vortex street at Re=1500)



Subsurface Flow





- **Multiscale problem**:
  - Very complicated geometry on mm scale
  - Impossible to solve Navier-Stokes
  - Pore-scale geometry depends on type of soil

Groundwater Flow

- Conservation of mass for an incompressible fluid

$$\nabla \cdot v(x) = f(x)$$

- **Darcy's law**

$$v(x) = -K(x)(\nabla p(x) - \rho \vec{g})$$

- $K(x)$ is given **permability**, $p(x)$ is unknown pressure, $\vec{g} = (0,0,-g)^T$
- Reformulate by setting $d(x) = -x_3$ and $\vec{g} = -g \nabla d(x)$, then

$$v(x) = -K(x)(\nabla p(x) + \rho g \nabla d(x)) = -K(x)\rho g \nabla \left(\frac{p(x)}{\rho g} + d(x) \right) = -k(x) \nabla H(x)$$

- with $k(x)$ **hydraulic conductivity** and $H(x)$ **hydraulic head**
- Combining mass conservation and Darcy's law gives

$$-\nabla \cdot (k(x) \nabla H(x)) = f(x)$$

Different Problems, same Math!

- Stationary heat conduction in a solid: $-\nabla \cdot (\lambda \nabla T) = f$
- Groundwater flow: $-\nabla \cdot (k \nabla H) = f$
- These seemingly different problems lead to the same PDE
- Write as system of PDEs:

$$\nabla \cdot q = f$$
$$q = -A \nabla u$$

- The first equation expresses conservation (heat energy, mass of water) **and is exact**
- The second equation is a flux model. **It is an approximation! The parameter function $A(x)$ (which may be matrix-valued) needs to be calibrated!**

Remarks

- The derivations in this presentation were informal
- Gauß theorem, Fundamental theorem of calculus of variations, interchanging differentiation and integration only hold under certain smoothness conditions on the functions involved.
- The required conditions were implicitly made
- Tomorrow we will be more specific about the functions involved

Summary and Outlook

- We have seen that PDEs are a major tool in mathematical physics
- With countless applications in fluid flow, solid mechanics, electromagnetics, ..
- Many mathematical questions now arise:
  - Do they always have a solution?
  - Under what assumptions do they have a solution?
  - Is the solution unique and stable?
  - How would we find a solution, if it exists?

DAY 2
Excursion to PDE Theory

Plan for the Lecture

- Introduce boundary conditions
- Get a grasp on PDE theory
- Understand well-posedness
- Know about classical types of PDEs
- Get to know the concept of weak solutions

Motivation

- We derived/mentioned several PDEs
  - Poisson's equation $\Delta\Psi = 4\pi\gamma\rho$
  - Heat equation $\partial_t(\rho c T) + \nabla \cdot (\rho c T v - \lambda \nabla T) + r\rho c T = f$
  - Euler equations, Navier-Stokes equations, groundwater flow, …
- These had different properties:
  - Linear vs. nonlinear, $\Delta\Psi_1 = f_1 \wedge \Delta\Psi_2 = f_2 \Rightarrow \Delta(\alpha_1\Psi_1 + \alpha_2\Psi_2) = \alpha_1 f_1 + \alpha_2 f_2$
  - Stationary vs. instationary
- Important questions now are
  - **Does a given PDE have a solution?**
  - **If yes, is it unique?**
  - **If yes, is it stable?**
  - **If yes, how can we determine it?**

Situation in Ordinary Differential Equations

- **Peano's existence theorem**: Let $f : [t_0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ be continuous. Then the initial value problem

$$\frac{dy}{dt}(t) = f(t, y(t)), \qquad y(t_0) = y_0,$$

  has a solution in $[t_0, t_0 + \delta)$

- **Picard-Lindelöf**: If f is Lipschitz-continuous in its second argument, the solution is unique and $\delta = T - t_0$
- **There is no such general theorem for partial differential equations!**
- **Statements are only possible for certain „types" (or classes) of equations**

Millenium Prize Problem on Navier-Stokes

Actually, PDE theory can be quite difficult. The following problem is one of the **seven millenium prize problems** of the Clay Institute of Mathematics announced on May 24, 2000

> *Prove or give a counter-example of the following statement*:
>
> In three space dimensions and time, given an initial velocity field, there exists a vector velocity and a scalar pressure field, which are both smooth and globally defined, that solve the Navier–Stokes equations.

Actually, OpenAI claims now that one of its AIs has found a counter example!

We begin much more modest with Poisson's equation …

What is a solution?

- Consider Poisson's equation in a **bounded domain** $\Omega \subset \mathbb{R}^3$

$$\Delta \Psi = 4\pi\gamma\rho \qquad \text{in } \Omega$$

- This means $\Psi(x_1, x_2, x_3)$ satisfies

$$\forall x \in \Omega : \quad \frac{\partial^2 \Psi}{\partial x_1^2}(x) + \frac{\partial^2 \Psi}{\partial x_2^2}(x) + \frac{\partial^2 \Psi}{\partial x_3^2}(x) = 4\pi\gamma\rho(x)$$

- We call this a **strong** solution (later there will be a **weak** solution)
- Here, Ψ must be at least twice continuously differentiable w.r.t. each variable x_1, x_2, x_3, shorthand $\Psi \in C^2(\Omega)$, to be a strong solution
- This also implies that the function ρ shold be continuous
- Note: Ω is an **open** set, e.g. $\Omega = \{x \in \mathbb{R}^3 : \|x\| < 1\}$
- The boundary $\partial\Omega = \{x \in \mathbb{R}^3 : \|x\| = 1\}$ is **not** included in the statement!

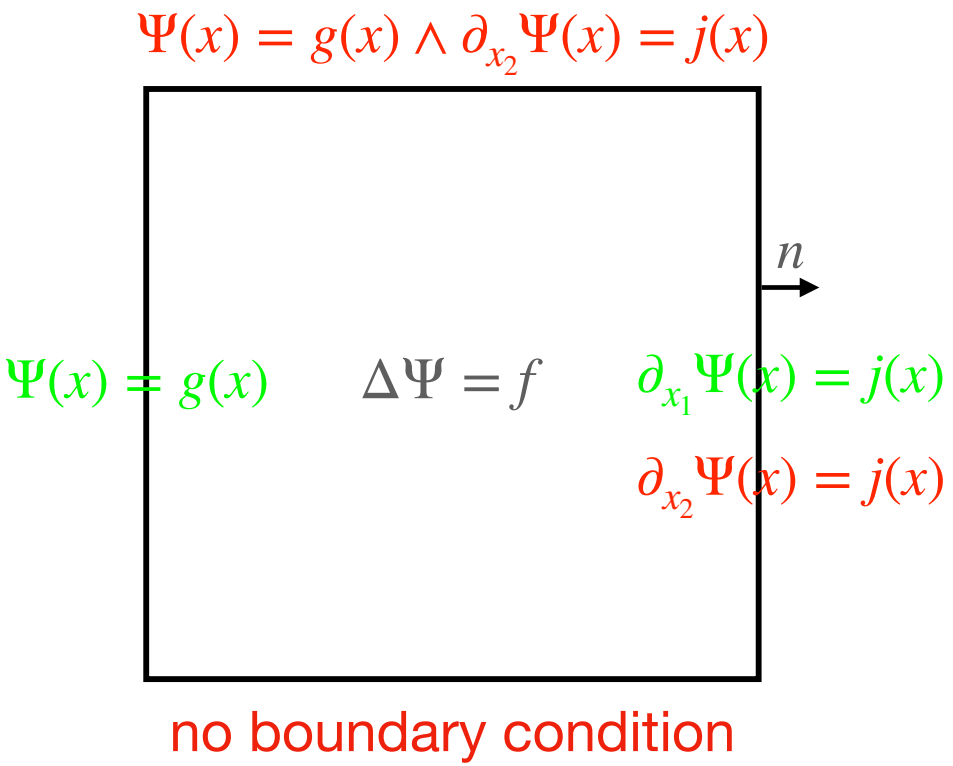
Boundary Conditions

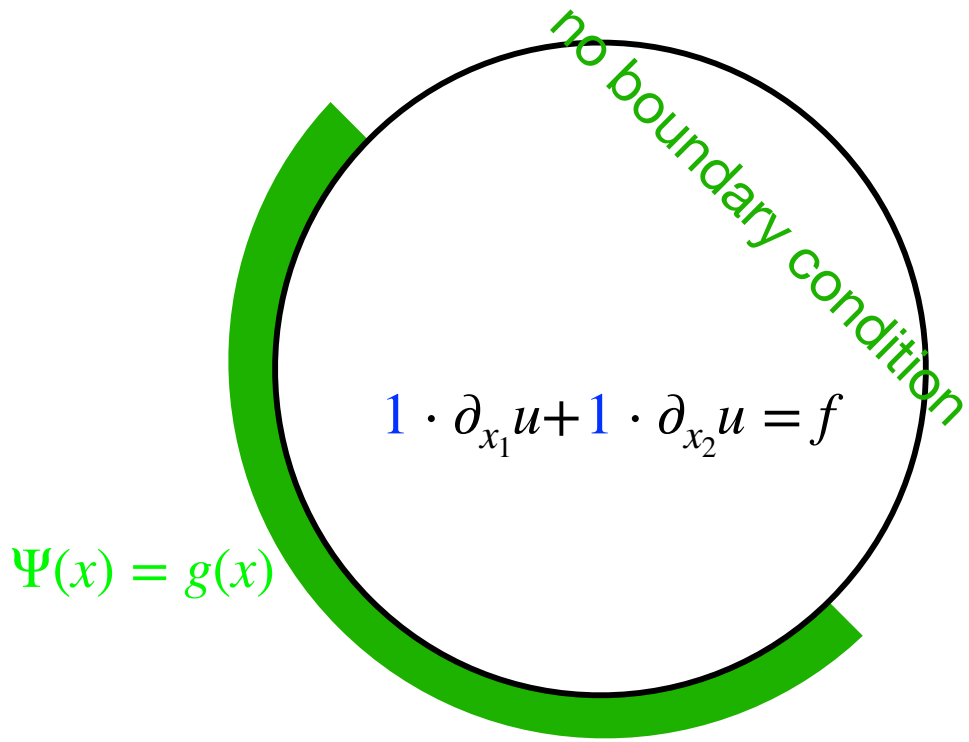
- What shall we do at the boundary, if we have a bounded domain?
- In short: **we need a boundary condition**! (Compare to initial condition in ordinary differential equations)
- Let Ψ be a solution of Poisson's equation $\Delta\Psi = 4\pi\gamma\rho$ in Ω. Then $\Psi + 1$ or $\Psi + 2x_1 - 3x_2 + 7$ are also solutions!
- Without boundary conditions, solutions of PDEs are often not unique
- Boundary conditions can ensure uniqueness of solutions
- Different boundary conditions are possible for Poisson's equation
  - Dirichlet condition: $\Psi(x) = g(x)$ for $x \in \partial\Omega$
  - Neumann condition: $\nabla\Psi(x) \cdot n(x) = j(x)$ for $x \in \partial\Omega$ (n: unit outer normal)
  - Mixtures of the two

Well-posedness

- A PDE problem is called **well-posed** in the sense of Hadamard if
  - **existence**: it has a solution
  - **uniqueness**: the solution is unique
  - **stability**: the solution depends continuously on the data (boundary conditions, parameters in the PDE, source term, …)
- Boundary conditions are often needed to ensure uniqueness
- The kind of boundary conditions and where they are posed is often determined by stability

Examples of Boundary Conditions





$$\partial\Omega_{in} = \{(x_1, x_2) \in \partial\Omega : 1 \cdot x_1 + 1 \cdot x_2 < 0\}$$

Type Classification

- **Second-order linear** PDE in two variables for $u : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$:

$$a_{11}\partial^2_{x_1x_1}u + 2a_{12}\partial^2_{x_1x_2}u + a_{22}\partial^2_{x_2x_2}u + b_1\partial_{x_1}u + b_2\partial_{x_2}u + c = f$$

- PDE is called
  - **elliptic**, if eigenvalues of $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$ are nonzero and have same sign
  - **hyperbolic**, if eigenvalues of A are nonzero and have opposite sign
  - **parabolic**, if **one** eigenvalue of A is zero and $\begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{12} & a_{22} & b_2 \end{pmatrix}$ has full rank
- Each type needs seperate treatment in theory and numerics

$$\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} = f$$

(Poisson, elliptic)

$$\frac{\partial^2 u}{\partial x_1^2} - \frac{\partial^2 u}{\partial x_2^2} = f$$

(Wave equation, hyperbolic)

$$\frac{\partial u}{\partial x_1} - \frac{\partial^2 u}{\partial x_2^2} = f$$

(Heat equation, parabolic)

Linear Second-Order Elliptic PDE

- From now on we concentrate on the **linear, second-order, elliptic** PDE

$$-\nabla \cdot (A \nabla u) = f \quad \text{in } \Omega \subseteq \mathbb{R}^d, \quad d = 2,3$$

  where $A \in \mathbb{R}^{d \times d}$ is symmetric and satisfying

$$a_{min} \leq x^T A x \leq a_{max} \qquad \forall x \in \mathbb{R}^d, x \neq 0$$

  for two numbers $0 < a_{min} < a_{max}$

- A is symmetric and positive definite, all its eigenvalues are real and positive
- The definition of ellipticity can so be extended to any dimension $d > 2$
- We add the **Dirichlet** and **Neumann** boundary conditions

$$u(x) = g(x) \text{ on } \Gamma_D \subset \partial\Omega, \qquad -(A \nabla u) \cdot n = j(x) \text{ on } \Gamma_N = \partial\Omega \backslash \Gamma_D$$

Weak Solution Concept (1)

- Assume that u now is a strong solution of $-\nabla \cdot (A \nabla u) = f$ in Ω
- Let $v \in C^1(\bar{\Omega})$ be an arbitrary function
- For any such function v use **integration by parts** (Green's 1st theorem):

$$-\int_\Omega \nabla \cdot (A \nabla u) v \, dx = \int_\Omega (A \nabla u) \cdot \nabla v \, dx + \int_{\partial\Omega} (-A \nabla u) \cdot n v \, ds = \int_\Omega f v \, dx$$

- The boundary integral term can be treated as follows:
  - Dirichlet boundary $x \in \Gamma_D$: choose $v(x) = 0$
  - Neumann boundary $x \in \Gamma_N$: insert boundary condition $-(A \nabla u) \cdot n = j(x)$

Weak Solution Concept (2)

- If u is a strong solution we get for any $v \in C^1(\bar{\Omega})$ with $v(x) = 0$ on Γ_D that

$$\int_\Omega (A \nabla u) \cdot \nabla v \, dx = \int_\Omega fv \, dx - \int_{\Gamma_N} jv \, ds$$

- The **test function** v acts as a filter to test the solution on $\omega = \operatorname{supp}(v) = \overline{\{x : v(x) \neq 0\}}$
- Now we change the direction of the argument: **If we choose a proper set of test functions V and require that**

$$\int_\Omega (A \nabla u) \cdot \nabla v \, dx = \int_\Omega fv \, dx - \int_{\Gamma_N} jv \, ds \qquad \forall v \in V$$

does this fix $u \in U$, from some appropriate set U, in a unique way?

- **Yes!** But what are appropriate U and V?

Lebesgue Space

- Measuring convergence of sequences of functions requires norms
- These norms should use integrals (because the weak form uses integrals)
- For $u \in C^0(\overline{\Omega})$ define the L^2**-norm**

$$\|u\|_{0,\Omega} = \sqrt{\int_\Omega u^2(x)\,dx}$$

- The space $C^0(\overline{\Omega})$ is, however, not complete w.r.t. this norm, i.e. limits of Cauchy sequences may not be elements of $C^0(\overline{\Omega})$
- Remedy: Define $L^2(\Omega)$ as the completion of $C^0(\overline{\Omega})$ w.r.t. $\|.\|_{0,\Omega}$
- This requires a proper definition of the Lebesgue-integral
- $\|u\|_{0,\Omega} = \sqrt{(u,u)_{0,\Omega}}$ with the L^2**-scalar product** $(u,v)_{0,\Omega} = \int_\Omega uv\,dx$
- $L^2(\Omega)$ is a **Hilbert space**, i.e. a normed, complete, linear space with scalar product

Sobolev Spaces

- The weak solution concept involves also derivatives under the integral
- For $u \in C^1(\overline{\Omega})$ define the H^1-scalar product

$$(u, v)_{1,\Omega} = \int_\Omega uv + \sum_{i=1}^{d} \partial_{x_i} u \, \partial_{x_i} v \, dx$$

- This induces the H^1-norm $\|u\|_{1,\Omega} = \sqrt{(u, u)_{1,\Omega}}$
- The Sobolev space $H^1(\Omega)$ is then the completion of $C^1(\overline{\Omega})$ w.r.t. $\| \cdot \|_{1,\Omega}$
- Requires also a proper extension of derivatives to Sobolev functions
- $H^k(\Omega)$, $k > 1$, is the Sobolev spaces involving k-th order derivatives
- $H^k(\Omega)$ is a Hilbert space
- $H_0^k(\Omega) \subset H^k(\overline{\Omega})$ is the subspace of functions which are zero on $\partial\Omega$ (almost everywhere)

Weak Formulation

- It turns out that $H^1(\Omega)$ and subspaces of it are the appropriate function spaces for the concept of weak solutions
- We make the following definitions

$$a(u,v) = \int_\Omega (A\nabla u)\cdot\nabla v\, dx, \qquad l(v) = \int_\Omega fv\, dx - \int_{\Gamma_N} jv\, ds$$

  $a(u,v)$ is a **bilinear form** and $l(v)$ is a **linear form**

- Then for the **homogeneous Dirichlet problem** $\Gamma_D = \partial\Omega$, $u(x) = 0$, $x \in \partial\Omega$, the **weak formulation** reads:

$$\text{Find } u \in H_0^1(\Omega) \text{ s. t.} \quad a(u,v) = l(v) \quad \forall v \in H_0^1(\Omega)$$

- Other boundary conditions require different subspaces of $H^1(\Omega)$

Lax-Milgram Theorem

Lax-Milgram theorem. Assumptions:

- V a real Hilbert space with norm $\|\cdot\|_V$ implied by scalar product $(\cdot\,,\cdot)_V$
- $l : V \to \mathbb{R}$ a continuous linear form, i.e. $l(v) \leq C_l \|v\|$
- $a : V \times V \to \mathbb{R}$ a continuous bilinear form, i.e. $a(u,v) \leq C_a \|u\|_V \|v\|_V$
- a is **coercive**, i.e. $\exists \alpha > 0 : \alpha \|v\|_V^2 \leq a(v,v)$ for all $v \in V$

Then there exists a unique $u \in V$ such that

$$a(u,v) = l(v) \quad \forall v \in V$$

Moreover,

$$\|u\|_V \leq C_l/\alpha$$

Remark: The bilinear form $a(u,v) = \int_\Omega (A\,\nabla u) \cdot \nabla v \, dx$ is coercive in $V = H_0^1(\Omega)$

Applications of Lax-Milgram

- LM can be used to show well-posedness of
  - elliptic PDE with Dirichlet, Neumann and mixed boundary condition
  - including a sink term such as $-\nabla \cdot (A \nabla u) + ku = f$ with $k \geq 0$
  - including a convective part as in $-\nabla \cdot (A \nabla u) + \beta \cdot \nabla u = f$
- Coercivity is a sufficient condition for well-posedness but not necessary
- Symmetry of the bilinear form is not required
- For the general case with Banach spaces U, V

$$\text{Find } u \in U \text{ s. t.} \quad a(u, v) = l(v) \quad \forall v \in V$$

  the Banach-Necas-Babuska theorem can be used
- It works in the parabolic case

Why is it Important (1)

- Proving well-posedness of strong solutions requires more assumptions on the data
- For example consider Poisson's equation $\Delta\Psi = 4\pi\gamma\rho$ let $\rho(x)$ be given by

$$\rho(x) = \begin{cases} 1 & \|x\| \leq R \\ 0 & \text{else} \end{cases}$$

- For a strong solution, discontinuous ρ is not allowed, it turns out not even simple continuity is sufficient
- For the weak solution, discontinuous ρ is perfectly fine. Only $\rho \in L^2(\Omega)$ is required
- One can also show: If the weak solution turns out to be in $C^2(\Omega) \cap C^0(\overline{\Omega})$ (for the Dirichlet problem), then it is also a strong solution

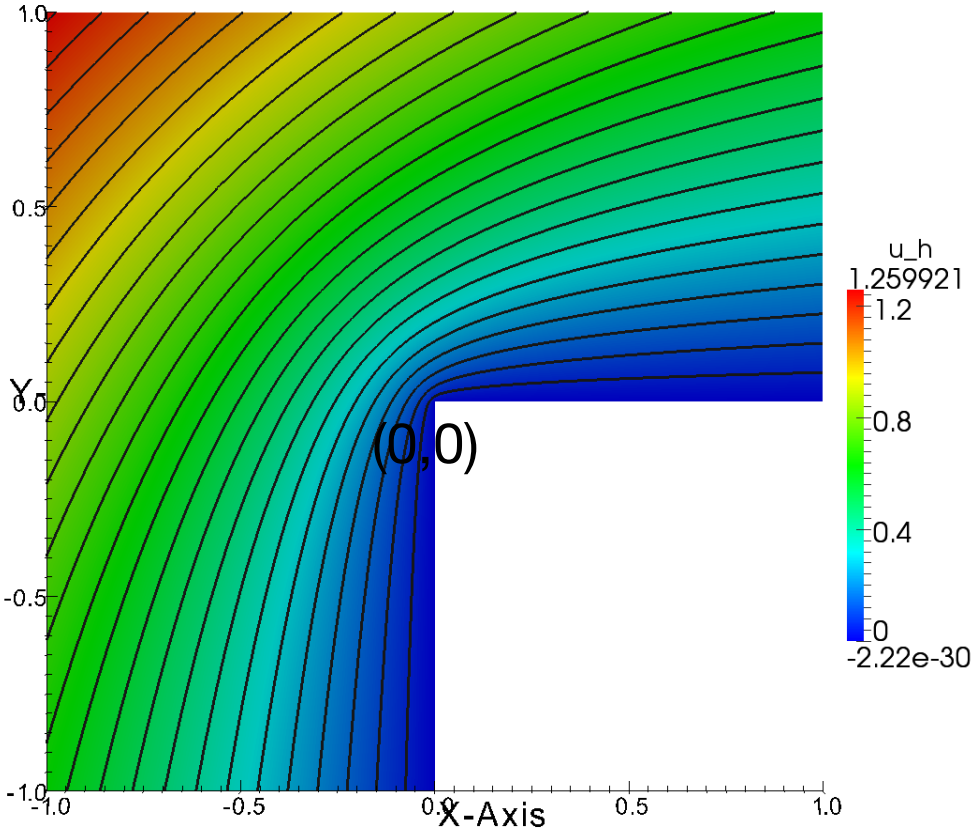
Reentrant Corner Problem

- Consider $\Delta u = 0$ in the domain $\Omega = (-1,1)^2 \backslash ([0,1) \times (-1,0])$
- One can show that

$$u(r,\theta) = r^{\frac{2}{3}} \sin\left(\frac{2}{3}\theta\right)$$

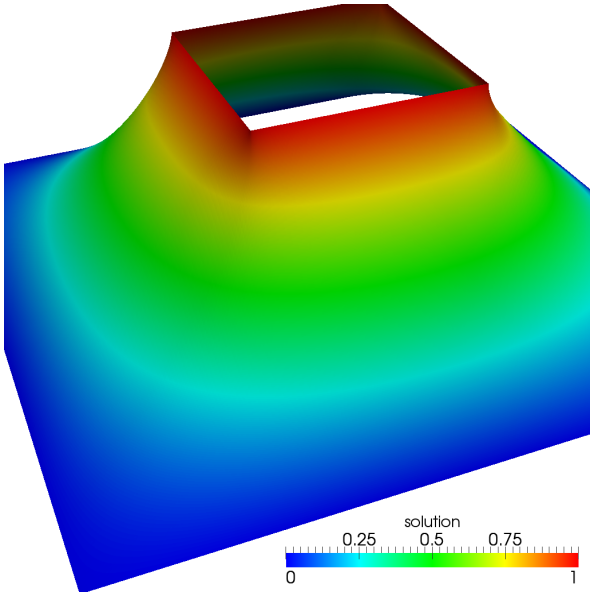
  is a solution expressed **in polar coordinates**

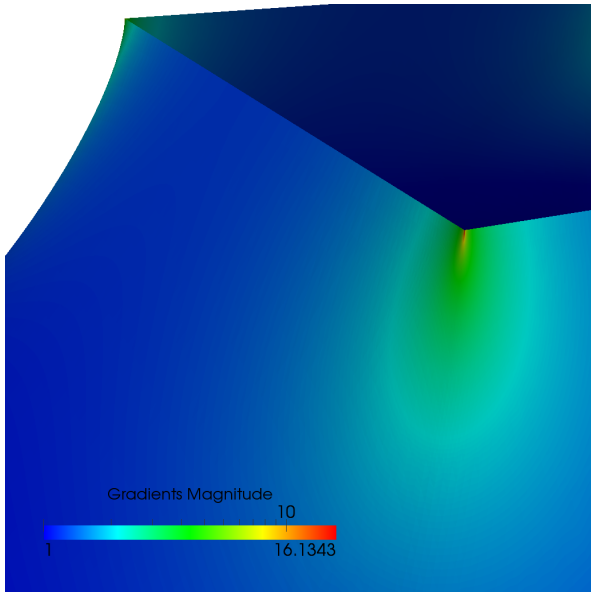
- $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ (strong solution!)
- $u = 0$ at $\{x \in \partial\Omega : \|x\| < 1\}$
- But $\lim_{r\to 0} \partial_r u(r,\theta) = \infty$
- u on $\partial\Omega$ near the origin is zero
- $u \in L^2(\Omega), u \in H^1(\Omega), u \notin H^2(\Omega)$



Reentrant Corner Problem (2)

- These „singular“ solutions are important in practice
- Application: Deformation of rubber sheet under its own weight fixed at the boundary
- Large forces near the corners in non-convex domains may lead to rupture
- It is challenging to compute such solutions accurately





Why is it Important (2)

- The weak solution concept is the basis of the **finite element method** (FEM), a highly successful numerical solution technique for elliptic (and other) PDEs
- FEM in a nutshell: Let $V_h \subset V$ be a **finite-dimensional subspace** in which the weak formulation is posed. E.g. $V_h \subset H_0^1(\Omega)$
- Lax-Milgram implies: $a(u_h, v) = l(v)\ \forall v \in V_h$ has a unique solution $u_h \in V_h$!
- **How can we compute u_h?**
- $V_h = \text{span}\{\varphi_1, \ldots, \varphi_N\}$ with **basis functions** φ_j because V_h has finite dimension
- Represent $u_h = \sum_{j=1}^{N} z_j \varphi_j$ and insert into weak formulation:

$$a\left(\sum_{j=1}^{n} z_j \varphi_j, v\right) = l(v)\ \forall v \in V_h \quad \Leftrightarrow \quad \sum_{j=1}^{n} z_j a\left(\varphi_j, \varphi_i\right) = l(\varphi_i)\ \forall i = 1, \ldots, n$$

$$\Leftrightarrow \quad \mathbf{Az} = \mathbf{b},\ (A)_{i,j} = a(\varphi_j, \varphi_i),\ (b)_i = l(\varphi_i)$$

Summary and Outlook

- There are different definitions of „solutions“ of a PDE
- Weak solutions exist under less stringent assumptions on the data
- Their definition requires tools from functional analysis
- Understanding weak solutions is helpful (but not necessary) for understanding the finite element method

DAY 3
Finite Element Method

Plan for the Lecture

- Get to know the Finite Element Method (FEM), a numerical solution technique
- Understand how the corresponding linear system $Az = b$ is set up
- Understand the convergence properties of FEM

Motivation

- Analytical solutions of PDEs require strong simplifications
- Numerical solutions can be computed for realistic geometries and data
- Several techniques exist (not exhaustive)
  - Finite Difference Method (FDM)
  - Finite Volume Method (FVM)
  - Finite Element Method (FEM)
- We will focus on FEM (today and FVM tomorrow). Why?
  - FEM is very flexible w.r.t. geometry and data of the PDE
  - FEM can handle linear/nonlinear, stationary/instationary, scalar/system cases
  - FEM has a well-developed convergence theory
  - Many open-source software packages for FEM/FVM are available

Recap the Linear Second-Order Elliptic PDE

- **Strong problem formulation:** focus on the homogeneous Dirichlet problem (other b.c. are possible)

$$-\nabla \cdot (A \nabla u) = f \quad \text{in } \Omega \subseteq \mathbb{R}^d, \quad d = 2,3$$
$$u = 0 \quad \text{on } \partial\Omega$$

- **Weak problem formulation** in $V = H_0^1(\Omega)$

$$\text{Find } u \in V : \underbrace{\int_\Omega (A \nabla u) \cdot \nabla v \, dx}_{a(u,v)} = \underbrace{\int_\Omega f v \, dx}_{l(v)} \quad \forall v \in V$$

- **Weak formulation in finite dimensional subspace** $V_h = \text{span}\{\varphi_1, \ldots, \varphi_N\} \subset V$
- Represent $u_h = \sum_{j=1}^N z_j \varphi_j$ and insert into weak formulation:

$$a\left(\sum_{j=1}^N z_j \varphi_j, v\right) = l(v) \ \forall v \in V_h \quad \Leftrightarrow \quad \sum_{j=1}^N z_j a\left(\varphi_j, \varphi_i\right) = l(\varphi_i) \ \forall i = 1, \ldots, N$$
$$\Leftrightarrow \quad \mathbf{Az} = \mathbf{b}, \ (A)_{i,j} = a(\varphi_j, \varphi_i), \ (b)_i = l(\varphi_i)$$

Basic Steps

- How to chose the finite element space V_h
- How to chose the basis functions φ_i?
- How to compute the required integrals?
- How to solve the linear system?
- We proceed in reverse order:
  - Computing integrals
  - Define basis functions
  - Define finite element space
  - Solving linear systems is postponed until Friday

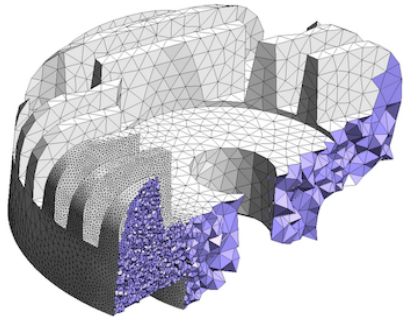
Computing Integrals

- So we need to compute the integrals

$$a(\varphi_j, \varphi_i) = \int_\Omega (A \nabla \varphi_j) \cdot \nabla \varphi_i \, dx, \qquad l(\varphi_i) = \int_\Omega f \varphi_i \, dx$$

- To accomplish this, we subdivide Ω into open subdomains ω_k of simple shape (triangles, quadrilaterals, tetrahedrons, hexahedrons, prisms, …)

$$\bigcup_{k=1}^{M} \overline{\omega}_k = \overline{\Omega}, \quad \omega_k \cap \omega_l = \emptyset \text{ if } k \neq l, \quad \mathbb{T}_h = \{\omega_1, \ldots, \omega_M\}$$



from gmsh.info

- This is the **mesh generation problem**
- Then $a(\varphi_j, \varphi_i) = \int_\Omega (A \nabla \varphi_j) \cdot \nabla \varphi_i \, dx = \sum_{k=1}^{M} \int_{\omega_k} (A \nabla \varphi_j) \cdot \nabla \varphi_i \, dx$

Excursion: Mesh Generation

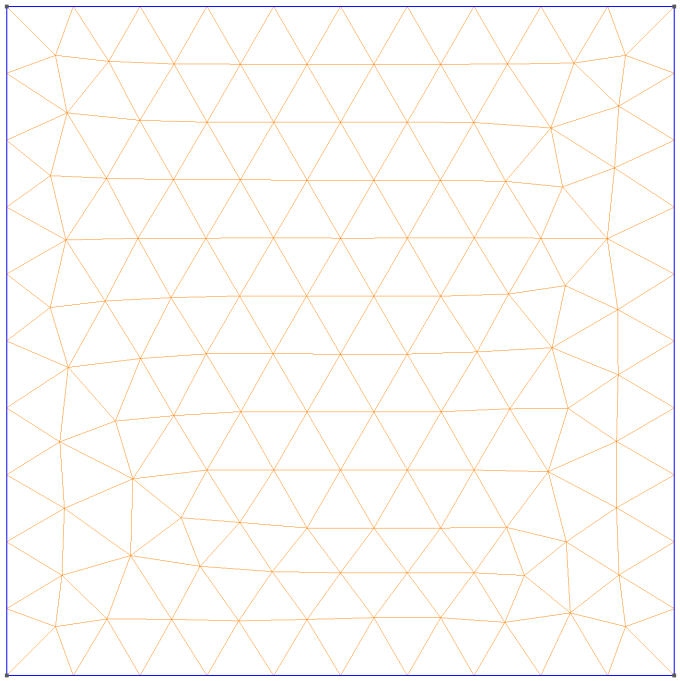
- Is a seperate and complicated problem by itself
- There is very good open-source software: **gmsh** https://gmsh.info/
- **Conforming Mesh**: Intersection of two triangles is a common edge, a common vertex or empty
- Description of the geometry through boundary representation (BREP)

```
c = 0.1;

Point(1) = {0, 0, 0, lc};
Point(2) = {1, 0, 0, lc};
Point(3) = {1, 1, 0, lc};
Point(4) = {0, 1, 0, lc};

Line(1) = {1,2};
Line(2) = {2,3};
Line(3) = {3,4};
Line(4) = {4,1};

Line Loop(100) = {1,2,3,4};
Plane Surface(200) = {100};
```



Introducing the Finite Element Space V_h

- There are many ways to introduce finite element functions!
- All of them use the finite element mesh
- „Bottom-up“ approach taken here:
  1. Introduce the „reference element“
  2. and (basis) functions on the reference element
  3. Transformation from reference element to general element
  4. Element-by-element combination

Basis Functions on the Reference Element

- Define the reference element (triangle) as

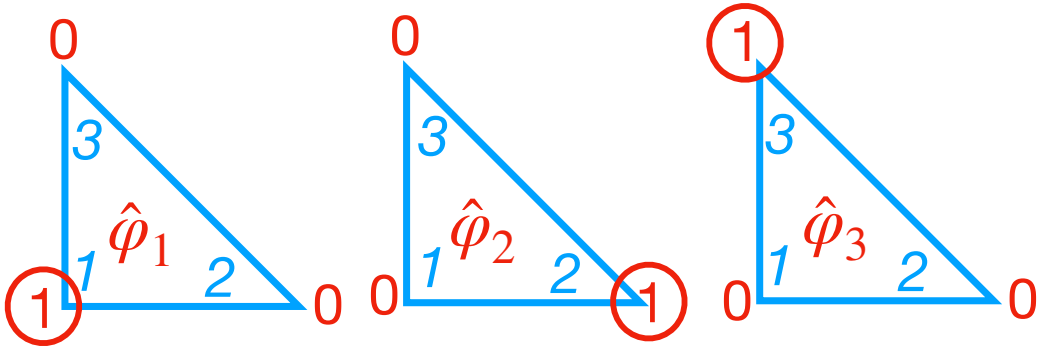
$$\hat{\omega} = \{(\hat{x}_1, \hat{x}_2) : \hat{x}_1 > 0 \wedge \hat{x}_2 > 0 \wedge \hat{x}_1 + \hat{x}_2 < 1\}$$

- Define basis functions on the reference element

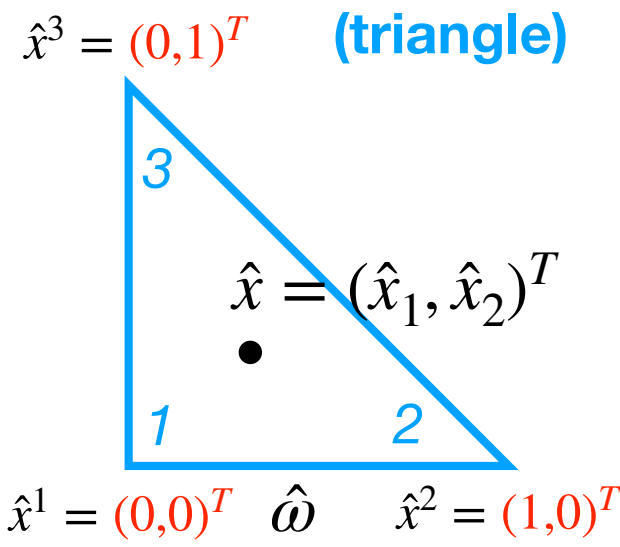
$$\hat{\varphi}_1(\hat{x}_1, \hat{x}_2) = 1 - \hat{x}_1 - \hat{x}_2, \quad \hat{\varphi}_2(\hat{x}_1, \hat{x}_2) = \hat{x}_1, \quad \hat{\varphi}_3(\hat{x}_1, \hat{x}_2) = \hat{x}_2$$

and observe

$$\hat{\varphi}_i(\hat{x}^j) = \begin{cases} 1 & i = j \\ 0 & \text{else} \end{cases}$$ (Kronecker property)



Reference element (triangle)



Transformation to General Element (1)

- Associate reference element with an arbitrary triangle
- **Local-to-global map** of vertex indices

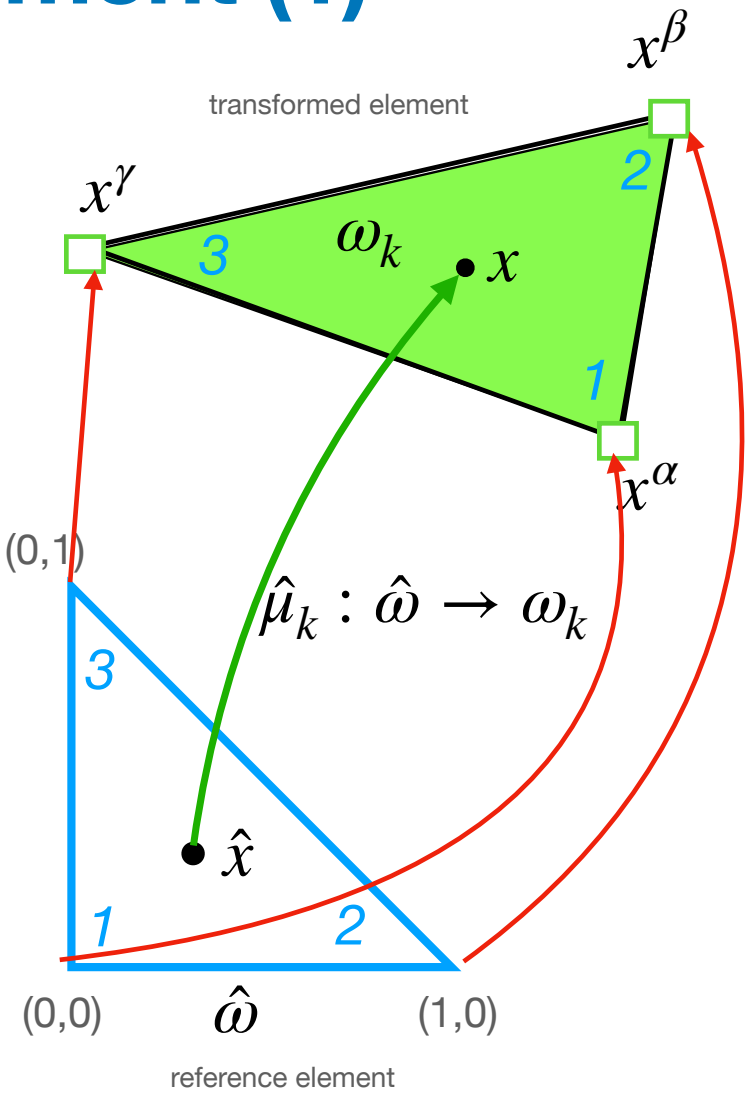
$$g(k,1) = \alpha, \quad g(k,2) = \beta, \quad g(k,3) = \gamma$$

- **Map** $\hat{\mu}_k : \hat{\omega}_k \to \omega_k$
- In case of triangles: $\hat{\mu}_k(\hat{x}) = B_k\hat{x} + b_k$, $B_k \in \mathbb{R}^{2\times 2}$:

$$B_k = \left[x^\beta - x^\alpha, x^\gamma - x^\alpha\right], \quad b_k = x^\alpha$$

- Defining basis functions on transformed element ω_k

$$\varphi_{g(k,i)}(x) = \hat{\varphi}_i(\underbrace{\hat{\mu}_k^{-1}(x)}_{\hat{x}}) \quad \Leftrightarrow \quad \varphi_{g(k,i)}(\underbrace{\hat{\mu}_k(\hat{x})}_{x}) = \hat{\varphi}_i(\hat{x})$$



Element-by-element Construction

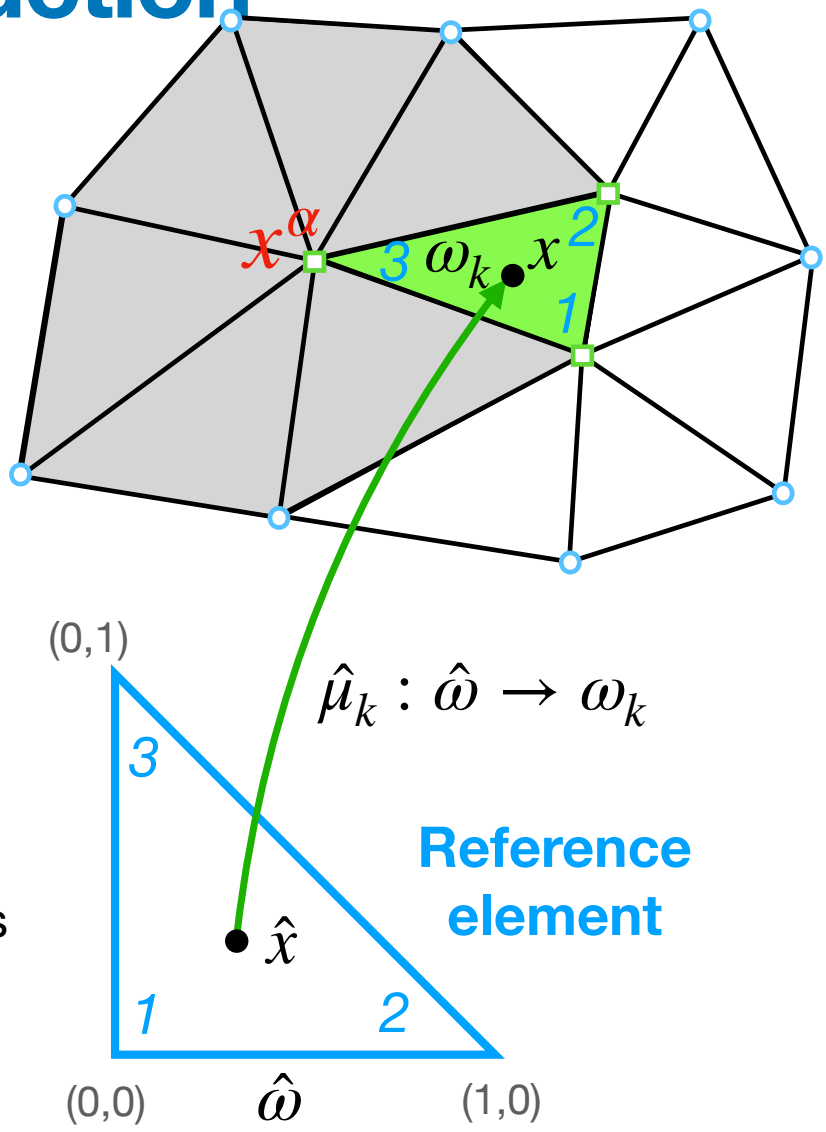
- „Global“ basis functions are defined element-by-element
- Consider φ_α corresponding to vertex α located at x^α

$$\varphi_\alpha(x) = \hat{\varphi}_i(\hat{x}) \text{ where } x = \mu_k(\hat{x}) \wedge g(k,i) = \alpha$$

- The global basis functions satisfy

$$\varphi_\alpha(x^\beta) = \begin{cases} 1 & \alpha = \beta \\ 0 & \text{else} \end{cases}$$

- Basis functions are nonzero only on very few elements, they have **local support**
- Basis functions φ_i are **continuous** (because the mesh is conforming)
- $V_h = \text{span}\{\varphi_1, \ldots, \varphi_N\}$, N is the number of **interior** vertices
- One can show $V_h \subset H_0^1(\Omega)$



Transformation to General Element (2)

- **Any function** $\hat{u}(\hat{x})$ on the reference element defines a function on the transformed element

$$u(x) = \hat{u}(\hat{\mu}_k^{-1}(x)) \quad \Leftrightarrow \quad u(\hat{\mu}_k(\hat{x})) = \hat{u}(\hat{x})$$

- Chain rule allows to compute gradients on the transformed element efficiently

$$\nabla u(\hat{\mu}_k(\hat{x})) = \hat{\nabla} \hat{\mu}_k(\hat{x})^{-T} \hat{\nabla} \hat{u}(\hat{x})$$

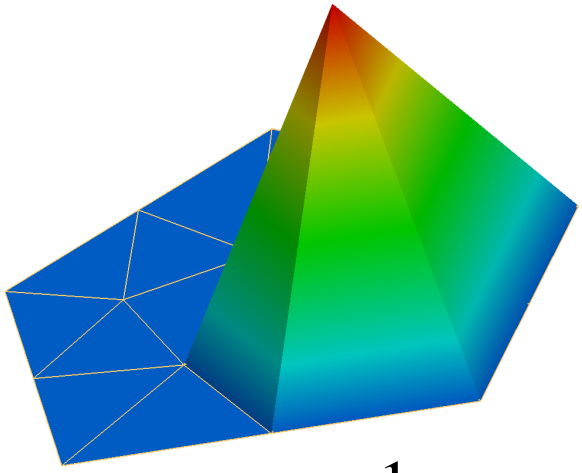
- Transformation formula for integrals

$$\int_{\omega_k} f(x)\, dx = \int_{\hat{\omega}} f(\hat{\mu}_k(\hat{x})) \,|\det \hat{\nabla} \hat{\mu}_k(\hat{x})|\, d\hat{x}$$

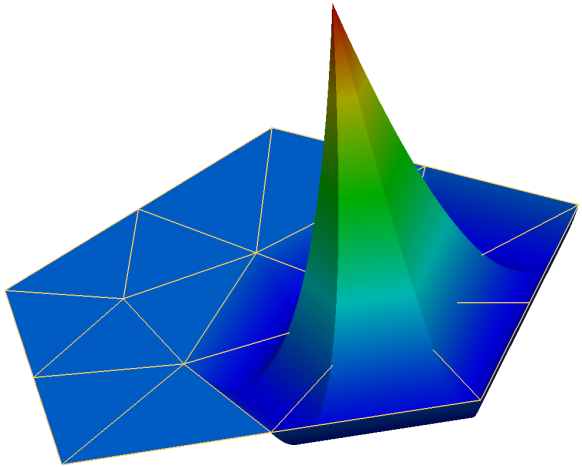
$$\int_{\omega_k} \nabla u(x) \cdot \nabla u(x)\, dx = \int_{\hat{\omega}} (\hat{\nabla} \hat{\mu}_k(\hat{x})^{-T} \hat{\nabla} \hat{u}(\hat{x})) \cdot (\hat{\nabla} \hat{\mu}_k(\hat{x})^{-T} \hat{\nabla} \hat{u}(\hat{x})) \,|\det \hat{\nabla} \hat{\mu}_k(\hat{x})|\, d\hat{x}$$

Generalization

- This procedure can be generalized to:
  - simplices of dimension $d > 2$
  - Other types of elements: cubes(quadrilaterals, hexahedra), prisms
  - Higher-order polynomials $p > 1$ per element
  - Elements with curved boundary ⇒ Objects with **smooth surfaces** can be handled
- **Very important**: $v \in V_h \subset H_0^1(\Omega)$ requires **continuity** of v, i.e. $v \in C^0(\bar{\Omega})$
- But: N increases exponentially in d („**curse of dimension**“)

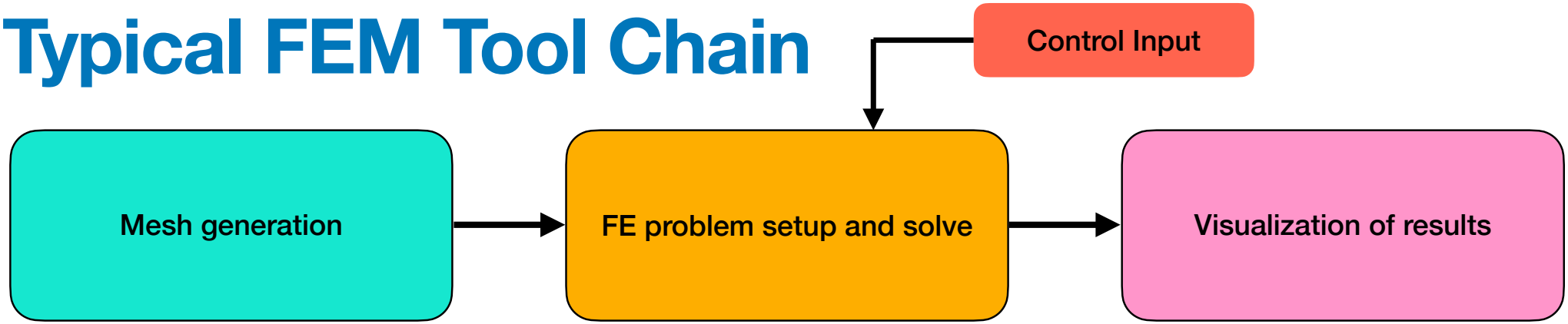


$p = 1$



$p = 2$

Typical FEM Tool Chain



- There are many high-quality open-source software packages:
- G**msh** https://gmsh.info/ for mesh generation
- General C++-based FE software packages (not exhaustive) :
  - **Dune** https://www.dune-project.org/
  - **deal.ii** https://dealii.org/
  - **MFEM** https://mfem.org/
  - **OpenFOAM** https://www.openfoam.com/
- **Paraview** https://www.paraview.org/ Visualization

Practical Session

- Show tool pipeline using Gmsh, Dune and Paraview
- Install Dune in seperate folder for that
  - /Users/peterbastian/Dune/dune-release211-26-07-24/release-build/dune-pdelab-tutorials/tutorial00/src

- Simple problem with polynomial solution
- Reentrant corner problem (2d)
- Gravity problem (3d)

Finite Element Error

- Weak solution and finite element solution satisfy:

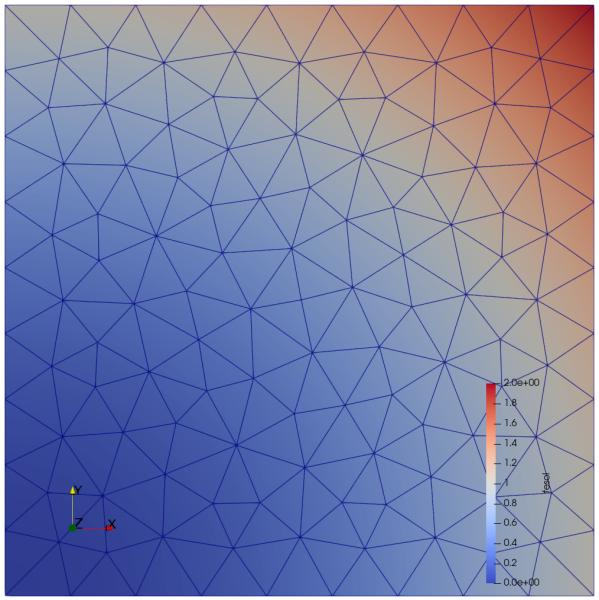
$$u \in H_0^1(\Omega): \qquad a(u,v) = l(v) \qquad \forall v \in H_0^1(\Omega)$$

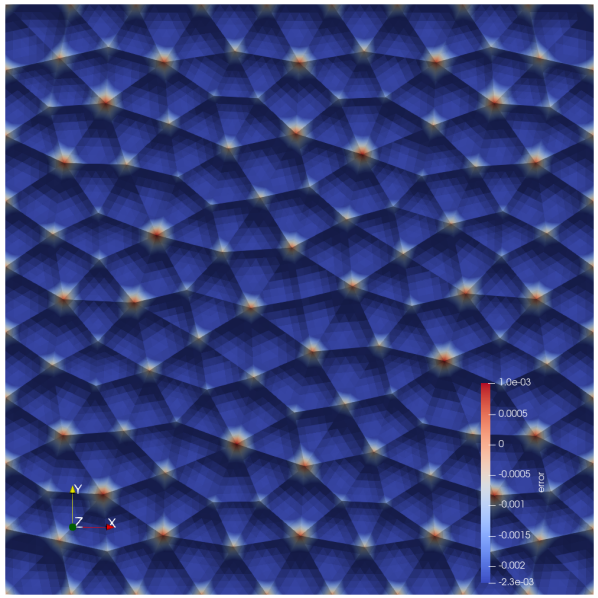
$$u_h \in V_h: \qquad a(u_h,v) = l(v) \qquad \forall v \in V_h$$

- Finite element error e is defined as

$$e = u - u_h$$

- Note: $e \in H_0^1(\Omega)$
- Top image shows piecewise linear finite element solution u_h
- Bottom image shows the error (visualized on a much finer mesh …)





A Priori Error Estimates (1)

- Starting point: Céa's lemma:

$$\|u - u_h\|_{1,\Omega} \leq \frac{C_a}{\alpha} \min_{v_h \in V_h} \|u - v_h\|_{1,\Omega}$$

  where α, C_a are constants from coercivity and continuity (see Lax-Milgram)

- $\min_{v_h \in V_h} \|u - v_h\|_{1,\Omega}$ is the best approximation in V_h measured in $\|.\|_{1,\Omega}$-norm
- For given u, choose the **finite element interpolant** $I_h(u) \in V_h$ satisfying

$$I_h(u)(x^j) = u(x^j) \quad \forall \text{ mesh vertices } x^j$$

- Then, clearly

$$\min_{v_h \in V_h} \|u - v_h\|_{1,\Omega} \leq \|u - I_h(u)\|_{1,\Omega}$$

A Priori Error Estimates (2)

- The definition of the finite element interpolant

$$I_h(u)(x^j) = u(x^j) \quad \forall \text{ mesh vertices } x^j$$

  requires $u \in H_0^2(\Omega)$ to be well-defined, $u \in H_0^1(\Omega)$ is not sufficient

- $u \in H_0^s(\Omega)$, for $\mathbb{N} \ni s > 1$ is called a **regularity assumption**
- **If** $u \in H_0^s(\Omega)$ and V_h is a finite element space with **polynomial degree** $p \geq 1$ and maximum element diameter h then there exists a constant (indep. of h) s.t.

$$\|u - u_h\|_{1,\Omega} \leq Ch^{\min(s-1,p)}$$

- This assumes „regular“ **refinement** of a given mesh
- Convergence order **depends on polynomial degree, regularity and norm**!

$$\|u - u_h\|_{0,\Omega} \leq Ch^{\min(s,p+1)}$$

Properties of the Linear System

- The finite element matrix A has many zero elements (depending on the mesh, dimension and polynomial degree)
- For fixed polynomial degree one typically has $\text{nonzeroes}(A) \leq CN$ under mesh refinement
- Asymptotically the finite element error decreases with increasing number of elements and therefore size of A
- Symmetry of the bilinear form $a(\cdot\,,\cdot)$ implies symmetry of A
- Coercivity of the bilinear form $a(\cdot\,,\cdot)$ implies positive definiteness of A
- Computational effort in FEM is dominated by the solution of the linear system

Computational Effort

Sparse direct solver (LU factorization)

Conjugate gradient method with incomplete LU factorization preconditioner

Conjugate gradient method with algebraic multigrid preconditioner

Number of unknowns

Create sparse matrix data structure

Assemble matrix entries (FEM)

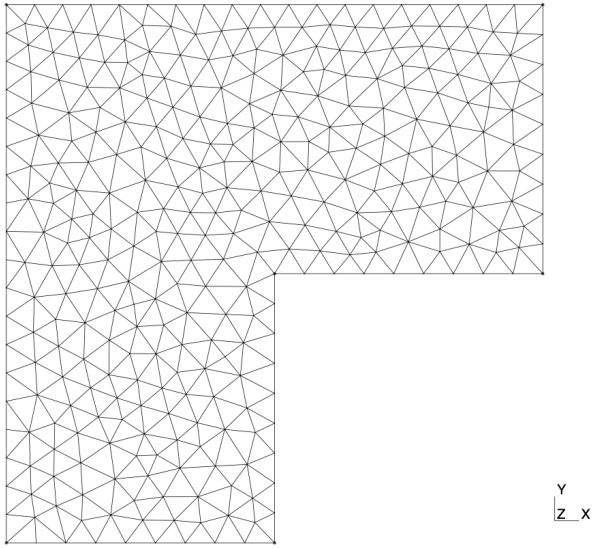
Space dimension

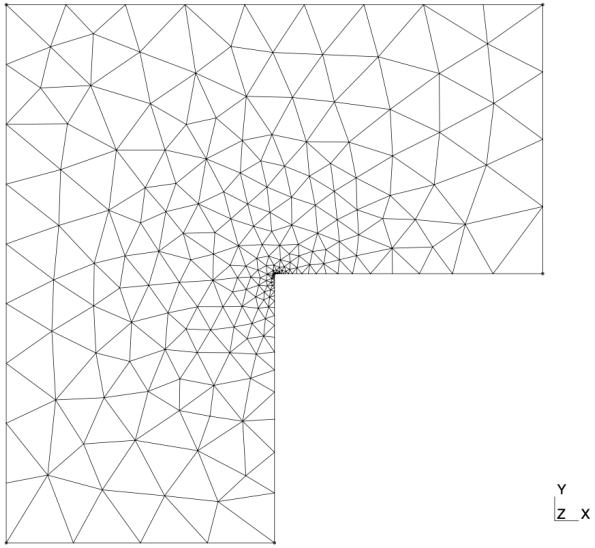
N	setup	assemble	UMFPack	CG ILU	CG AMG
			$d = 2$		
425	0.0007	0.0005	0.0008	0.0012	0.0025
1603	0.0009	0.0009	0.0042	0.0030	0.0064
6215	0.0031	0.0027	0.0168	0.0184	0.0216
24463	0.0183	0.0104	0.0702	0.0919	0.0699
97055	0.0476	0.0397	0.3171	0.3872	0.1933
386623	0.1873	0.1617	1.6884	3.0696	0.8291
1543295	0.7776	0.6605	9.1513	24.6798	3.5930
6166783	3.1583	2.6798	60.9960	191.2690	15.9926
			$d = 3$		
141	0.0002	0.0003	0.0003	0.0015	0.0005
785	0.0018	0.0023	0.0026	0.0045	0.0052
5081	0.0189	0.0180	0.0368	0.0255	0.0353
36177	0.0965	0.1356	0.8597	0.1885	0.2215
272161	0.7306	1.0814	34.3674	1.8256	1.6304
2109505	5.7102	8.9893	-	19.1134	13.7363

- Poisson equation with P1 finite elements
- Run-time on a single core Apple M2 Max in seconds

Adaptive Mesh Refinement

- One way to speed up the FEM is to adapt the mesh according to the solution u





N	$\|u-u_h\|_{0,\Omega}$	L^2-rate	$\|u-u_h\|_{1,\Omega}$	H^1-rate
$k=1$, uniform mesh				
334	4.8614e-03		1.0345e-01	
1259	1.9469e-03	1.3202	6.5916e-02	0.65024
4885	7.7909e-04	1.3213	4.1791e-02	0.65743
19241	3.1151e-04	1.3225	2.6431e-02	0.66099
76369	1.2438e-04	1.3245	1.6692e-02	0.66307
304289	4.9596e-05	1.3265	1.0532e-02	0.66439
$k=1$, adapted mesh				
250	1.4174e-03		5.9661e-02	
919	3.8599e-04	1.8766	3.1300e-02	0.93063
3517	1.0591e-04	1.8657	1.6561e-02	0.91836
13753	3.0263e-05	1.8072	8.9265e-03	0.89164
54385	9.1965e-06	1.7184	4.9287e-03	0.85688
216289	2.9978e-06	1.6172	2.7940e-03	0.81891